

MAZE SOLVER ROBOT - COMPLETE WIRING REFERENCE SHEET

1. Board & Control Inputs

Component	Function / Details	Connection Target
Board	STM32 Black Pill Development Board	F401CCU6
Push Button 1	User Input 1	PA2
Push Button 2	User Input 2	PA3

2. Motor Driver Logic (TB6612FNG <-> STM32)

TB6612FNG Pin	Signal Function	STM32 Pin Connection
PWMA	Left Motor PWM Speed Control	PA8 (TIM1_CH1)
AIN1	Left Motor Direction Control 1	PB0
AIN2	Left Motor Direction Control 2	PB1
PWMB	Right Motor PWM Speed Control	PA9 (TIM1_CH2)
BIN1	Right Motor Direction Control 1	PB2
BIN2	Right Motor Direction Control 2	PB10
STBY	Driver Standby Control (Active High)	PB12
VCC	Logic Power Input	5V (Buck Converter)
GND	Logic Ground	GND (Common Ground)

3. Motor Outputs & Terminals

TB6612FNG Output	Motor Connection Terminal	Details
BO1	Left Motor Terminal 2 (Red / M2)	Left Motor Drive Output
BO2	Left Motor Terminal 1 (White / M1)	Left Motor Drive Output
AO1	Right Motor Terminal 2 (Red / M2)	Right Motor Drive Output
AO2	Right Motor Terminal 1 (White / M1)	Right Motor Drive Output

4. Quadrature Encoders

Encoder Wire	Signal Name	Left Motor Connection	Right Motor Connection
Black	Encoder VCC (Power)	3.3V (AMS1117)	3.3V (AMS1117)
Blue	Encoder GND (Ground)	GND	GND
Green	Channel 1 (C1 Output)	PA6 (TIM3_CH1)	PA0 (TIM2_CH1)
Yellow	Channel 2 (C2 Output)	PA7 (TIM3_CH2)	PA1 (TIM2_CH2)

5. Time-of-Flight Sensors (I2C Array)

Sensor Target	SDA Pin	SCL Pin	XSHUT GPIO Pin	Power Supply VCC
Left ToF	PB9 (I2C1)	PB8 (I2C1)	PB7	3.3V (AMS1117)

Front Left ToF	PB9 (I2C1)	PB8 (I2C1)	PB6	3.3V (AMS1117)
Center ToF	PB9 (I2C1)	PB8 (I2C1)	PA15	3.3V (AMS1117)
Front Right ToF	PB9 (I2C1)	PB8 (I2C1)	PA4	3.3V (AMS1117)
Right ToF	PB9 (I2C1)	PB8 (I2C1)	PA10	3.3V (AMS1117)

6. Power Regulation & Distribution

Source Voltage	Regulation Block / Destination	Output/Target Pin	Wiring Purpose
12V Battery	Direct Connection to TB6612FNG VM	VM Pin	High-power motor drive current
12V Battery	5V Step-Down Buck Converter Input	5V Buck Input	Main logic system supply voltage
5V Buck Output	STM32 Logic VCC	5V / VCC Pin	Powering the microcontroller board
5V Buck Output	TB6612FNG VCC	VCC Pin	Powering the motor driver logic gates
5V Buck Output	AMS1117 LDO Regulator Input	VIN Pin	Step-down to stable 3.3V supply
3.3V AMS1117 Out	5x Time-of-Flight Sensors	VCC Pins	Low-noise supply for ToF sensor array
3.3V AMS1117 Out	2x N20 Motor Quadrature Encoders	VCC Pins (Black Wire)	Stable voltage for magnetic hall sensors
Common Ground	All ground paths tied together	GND Pins	Shared signal and power return paths